Fiscal Year:	FY 2005	**Task Last Updated:**	FY 12/19/2007
PI Name:	Kennedy, Ann R. D.Sc.		
Project Title:	Countermeasures for Space Radiation Induced Myeloid Leukemia		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Immunology, Infection and Hematology Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	19104-6072	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
Start Date:	06/01/2004	**End Date:**	05/31/2007
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Gewirtz, Alan (University of Pennsylvania) Biaglow, John (University of Pennsylvania) Ware, Jeffrey (University of Pennsylvania) Stewart, Jelena (University of Pennsylvania) Newberne, Paul (Boston Medical Center)		
Grant/Contract No.:	NCC 9-58-IIH00402		
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Task Description:	This proposed research investigates mechanisms to resist malignancy in space with an emphasis on evaluation of certain nutritional supplements as countermeasures to protect against space radiation-induced leukemogenesis; these nutritional supplements include L-selenomethionine (SeM), vitamin C, vitamin E succinate, alpha-lipoic acid, Co-enzyme Q10 and N-acetylcysteine. Data obtained from our previous in vitro and animal studies supported by the National Space Biomedical Research Institute (NSBRI) have demonstrated that the nutritional supplements to be evaluated in this proposal are effective in preventing radiation induced oxidative stress and protecting cultured human cells against adverse biological effects induced by the types of radiation of most concern during space travel, i.e., highly energetic heavy charged particles (known as HZE particles) and protons. This proposed investigation will determine the efficacy of a combination of these nutritional supplements as a countermeasure against space radiation induced acute myeloid leukemia (AML) using the CBA mouse model system. Two 2-year animal experiments are planned to determine the effect of supplement combination on the development of AML in CBA mice exposed to radiation with HZE particles (1-GeV or 5-GeV/nucleon iron ions) or protons; blood cell profiles and gene expression (of myeloperoxidase [MPO] in the spleen and CD33 in the bone marrow) patterns will be monitored throughout the studies, and gene expression patterns of the oxidation resistance gene (OXR1) will be determined at the end of the studies. In addition, several short-term animal and cell culture experiments will be performed to determine the effects of radiation treatment and nutritional supplementation on three selected surrogate endpoint biomarkers (SEBs), which are the host bio-reduction capacity measured as plasma total antioxidant status, plasma protein carbonyl content and OXR1 gene expression in selected populations of white blood cells. The results of this proposed study are expected to provide critical information about the feasibility and mechanism(s) of nutritional supplements in increasing the resistance to space radiation induced malignancy. In the past year, we have observed that the dietary supplement countermeasures at the levels to be used in long-term studies are not toxic to CBA mice. In addition, CBA mice have been exposed to protons or to iron ions, with or without supplementation with the dietary nutritional supplements, for the long-term AML induction experiments. In the next year of the grant, animals will continue to be monitored as part of the long-term animal studies described in the grant and more of the short-term animal and cell culture experiments will be performed.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	People living on earth are exposed to higher than normal doses of radiation from a variety of sources and for a variety of reasons. It is expected that the countermeasures being developed as part of this NSBRI research project will be equally useful for the protection of people on earth from the expected adverse effects of exposure to ionizing radiation.
Task Progress:	We have observed that the antioxidant combination dietary supplement is non-toxic to CBA mice, so the leukemogenesis studies have been started with animal exposures to radiation from iron ions or protons. In these studies, the animals are maintained on diets with or without the dietary supplement cancer preventive agents throughout the studies. These long-term animal studies will continue throughout the coming year. Our mechanistic studies of the past year have focused on the Oxidation Resistance 1 (OXR1) gene, which is thought to play a role in cellular protection against oxidative stress. One of the aims of our grant is to investigate whether the expression of OXR1 is increased in mononuclear cells exposed to ionizing radiation from HZE particles. In our experiments, regulation of OXR1 gene expression was studied in vivo in a mononuclear cell population taken from CBA mice and in vitro in CRL-9855 cells, a line of human mononuclear cells. We used quantitative PCR to evaluate the effects of chemopreventive agents, selenomethionine (L-SeM) alone or in combination with several antioxidants, as well as the effects of ionizing radiation from HZE particles on OXR1 gene expression. We have found that both chemopreventive agents and HZE particle radiation have an effect on the transcriptional regulation of OXR1 gene expression. CBA mice (6 per group) were fed the agent supplemented diets 3 days prior to the radiation exposure. Mice were exposed to a single dose of 0 or 50 cGy from iron ions, and 24 hours post radiation exposure mononuclear cells were isolated from mouse whole blood. Total RNA was isolated from the mononuclear cells and analyzed for expression of OXR1 using a dual labeled TaqMan probe and an ABI Prism 7700 setup. Both the SeM supplemented diet and the antioxidant combination diet led to increased expression of OXR1 in the mouse monocyte population. Exposure to radiation alone did not significantly affect expression of OXR1. However, mononuclear cells from irradiated mice fed the agent supplemented diets, showed significantly elevated expression of OXR1 compared to the untreated control group. In our in vitro studies, CRL-9855 cells were maintained in the agent supplemented medium 24 hours prior to the radiation exposure. The cell suspension was exposed to a single dose of 0 or 10 cGy from iron ions, and 24 hours post-irradiation total RNA was isolated for the OXR1 analysis. Supplementation with either SeM alone or the supplement combination resulted in significantly increased expression of OXR1 in the human monocyte cells. Exposure to HZE radiation in the presence or absence of chemopreventive agents significantly affected expression of OXR1. Expression of OXR1 was significantly elevated in all irradiated samples compared to the untreated control group.
Bibliography Type:	Description: (Last Updated: 08/24/2020)
Awards	Kennedy A. " NASA Spotlighted Investigator (October 2004). " Oct-2004
Presentation	Guan, J., Stewart, J. A., Ware, J. H. and A. R. Kennedy "The Effect of Dietary Supplement Agents on Radiation Induced Oxidative Stress in CBA Mice " Mar-2005
Presentation	Kennedy, A. R. "Countermeasures for the Biological Effects of Space Radiation " Mar-2005